

Revisiting Generalized Models of Japanese Seafood Demand: A Forecast Combination Approach

Susan Paudel

Department of Agriculture and Applied Economics, University of Georgia
susan.paudel@uga.edu

A. Ford Ramsey, Ph.D.

Department of Agriculture and Applied Economics, University of Georgia

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Abstract

We revisit the generalized inverse and ordinary demand systems for fish using four decades of monthly Japanese seafood data (1985–2025), a period over which the market shifted from domestic catch to heavy import dependence and reduced consumption. The earlier consensus no longer holds: neither the inverse nor the ordinary specification is decisively preferred. Faced with the model selection uncertainty, we examine whether combining the competing forecasts is preferable to selecting one. Using unconditional rolling-window forecasts and a range of combination methods, we find that no single model forecasts beats all others at every horizon. But the best combination forecasts are better than the best individual models forecasts. Gains from combination are modest because the eight nested demand systems produce highly similar forecasts. Forecast combination offers a reliable security against choosing the wrong model, even when it does not hugely improve on the best one.

Keywords: forecast combination; inverse demand systems; ordinary demand systems; Japanese seafood demand; model uncertainty; structural change

1 Introduction

Japan has long been a major producer, consumer, and trader of fish and seafood (Kamoey et al., 2015). Japan was the largest seafood producer, with peak production in 1984, producing 12.8 million tonnes but has been declining since (Food and Agriculture Organization (FAO), 2014). The seafood production decline has been exacerbated with calamities like 2011 Great East Earthquake and the following tsunami and nuclear accident in Fukushima which caused severe damage to fishing industry (Kamoey et al., 2015). Furthermore, eating habits of Japanese regarding seafood is also changing as compared to heavily relying on seafood in 1990s. While global demand of seafood is growing, in Japan the demand is declining with per capita meat consumption overtaking per capita fish consumption in 2011 (Kamoey et al., 2015).

In the 1980s, most of Japan's seafood in the consumption market came from domestic wild catch, so supply depended on natural factors like fish stocks and weather, not (Nakamoto, 2000). However, the structure of Japan's seafood market has changed significantly since the late 1980s. The share of seafood from domestic harvests dropped from 86% in 1985 to 60% by the late 1990s (Nakamoto, 2000), and Japan now relies heavily on imports, which peaked at 3.82 million tonnes in 2001 (Kamoey et al., 2015). While fresh fish are perishable and thus need to be consumed causing price to adjust, frozen fish are storable and can be modelled as quantity adjusting to price (Eales et al., 1997). Evolution of seafood freezing technology over years further illudes how the household preferences have evolved (**jia2022icy**). Given the changing supply structure and technology it's necessary to reevaluate the inverse and ordinary demand models for modeling Japanese household seafood consumption pattern.

In this paper we try to address the model selection uncertainty in household seafood demand. We use the updated version of monthly time series data on household demand and prices from 1985 to 2025 previously used by Eales et al. (1997). We begin with retesting the demand models (ordinary and inverse) (Eales et al., 1997). We then move on to out-of-sample forecasting and

evaluating the models based on out-of-sample forecast performance and finally check if forecast combination adds any value in the demand model selection.

Model selection concern has been addressed previously by specifying and estimating multiple models independently (Bryant & Davis, 2008). The early research about Japanese seafood demand was based on the assumption of demand are price driven for the market of 1980s (Wessells & Wilen, 1994). Eales et al. (1997) formulated inverse demand model with assumption of quantity driving the demand for fresh fish and ordinary demand model with assumption of price driving the demand for frozen fish. The empirical test identified inverse demand model better described the Japanese household preference on seafood for the market of 1980s. Tonsor and Marsh (2007) found out the generalized almost ideal demand system is preferred to the almost ideal demand system for Japanese household fish demand based on in-sample and out-of-sample performance. Bryant and Davis (2008) uses Bayesian averaging of estimates approach in comparison to classical pairwise testing procedure to account for model uncertainty in multiple dimension simultaneously. While Bayesian model averaging uses in-sample fit, Klaiber and Holt (2010) suggested examining demand models in terms of post-sample forecasting allows model to answer policy question that lie beyond the range of the data.

This study makes three contributions to the literature. First, the framework of Eales et al. (1997) is revisited and extended to contemporary data, providing empirical evidence that the historical dominance of inverse demand systems in Japanese seafood market no longer holds. Unlike in the 1990s, the current data do not decisively favor either the inverse or the ordinary specification, and the instruments that once identified these systems have weakened. Second, classical procedures have explored the demand models either in-sample modelling or conditional forecasting. We extend the evaluation of generalized household demand models through out-of-sample unconditional forecasting performance as used by Klaiber and Holt (2010) for seafood import demand. Finally, prior studies compared competing demand specifications with the goal of selecting the best one.

We instead treat model selection between inverse and ordinary specifications as a source of forecast uncertainty and apply the forecast combination method of Ramsey and Adjemian (2024) to a pool that includes both families. To our knowledge, this is the first application of forecast combination to a mixed inverse-and-ordinary demand system pool.

The remainder of the article is organized as follows. Section 2 describes the transition of the Japanese seafood market from domestic catch to imports and the changes in the seafood demand. Section 3 outlines the data used for analysis. Section 4 presents the generalized ordinary and inverse demand models and details the forecast combination framework adopted to address model uncertainty. Sections 5 provide the empirical results and Section 6 concludes with implications for demand estimation.

2 Background

Japan's seafood market has gone through concurrent supply side changes as well as demand side changes since the late 1980s. On the supply side, the source of Japanese seafood has shifted from domestic catch to imports. Figure 1 plots the composition of supply using data from FAOSTAT. Domestic production peaked in 1984 and has declined steadily since, falling to 4.8 million tonnes by 2012, with imports rising to bridge the gap and reaching a peak of 3.82 million tonnes in 2001 (Kamoey et al., 2015). In the 1980s, when supply came largely from domestic harvest, the quantity available was effectively fixed in the short run by fish stocks and weather; price adjusted to clear the market, and quantity could be treated as predetermined—the condition under which an inverse demand system is the natural specification. As imports came to supply a large and flexible share of consumption, that biological constraint loosened: quantity can now respond to price, which is the condition favoring an ordinary specification. The supply transition weakens the assumption for inverse demand model previously tested in Eales et al. (1997).

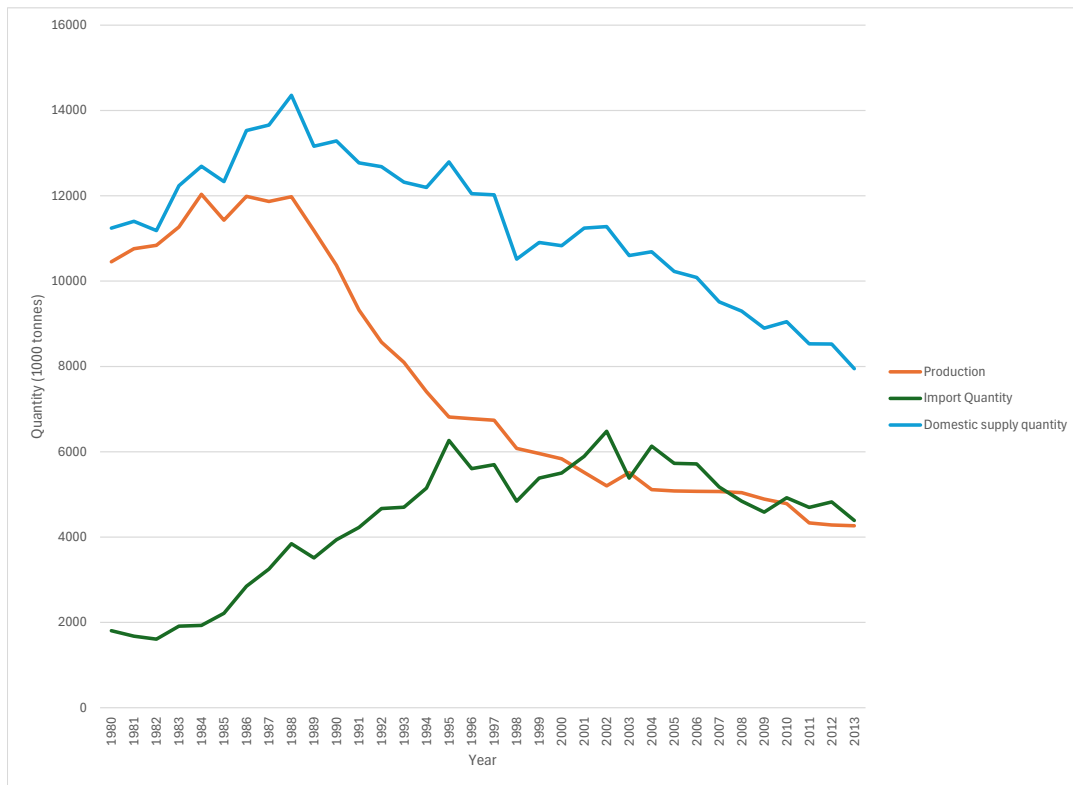


Figure 1: Domestic Production vs. Imports of Fishery Products in Japan (1980–2013) (source: FAOSTAT)

A simultaneous change has occurred on the demand side. Figure 2 plots per capita seafood and meat consumption from the late 1980s. Seafood consumption was stable until the early 2000s and has fallen since, from a peak of roughly 40 kg to 21.4 kg in 2023, while meat consumption rose past seafood in 2011 and has continued to climb. This is not only a response to relative prices but a shift in preferences. Younger consumers think fish is harder to prepare than meat, reflecting a shift to Western diets (Aruga & Wakamatsu, 2024; Shinsugi & Takimoto, 2025; Smil & Kobayashi, 2012). The market now has older consumers who prefer traditional fish dishes and younger ones who like easy, meat-like options like boneless or pre-cooked fish (Sasaki et al., 2022; Shinsugi & Takimoto,

2025). As consumers increasingly substitute meat for fish, the demand for fish has become much more sensitive to price changes.

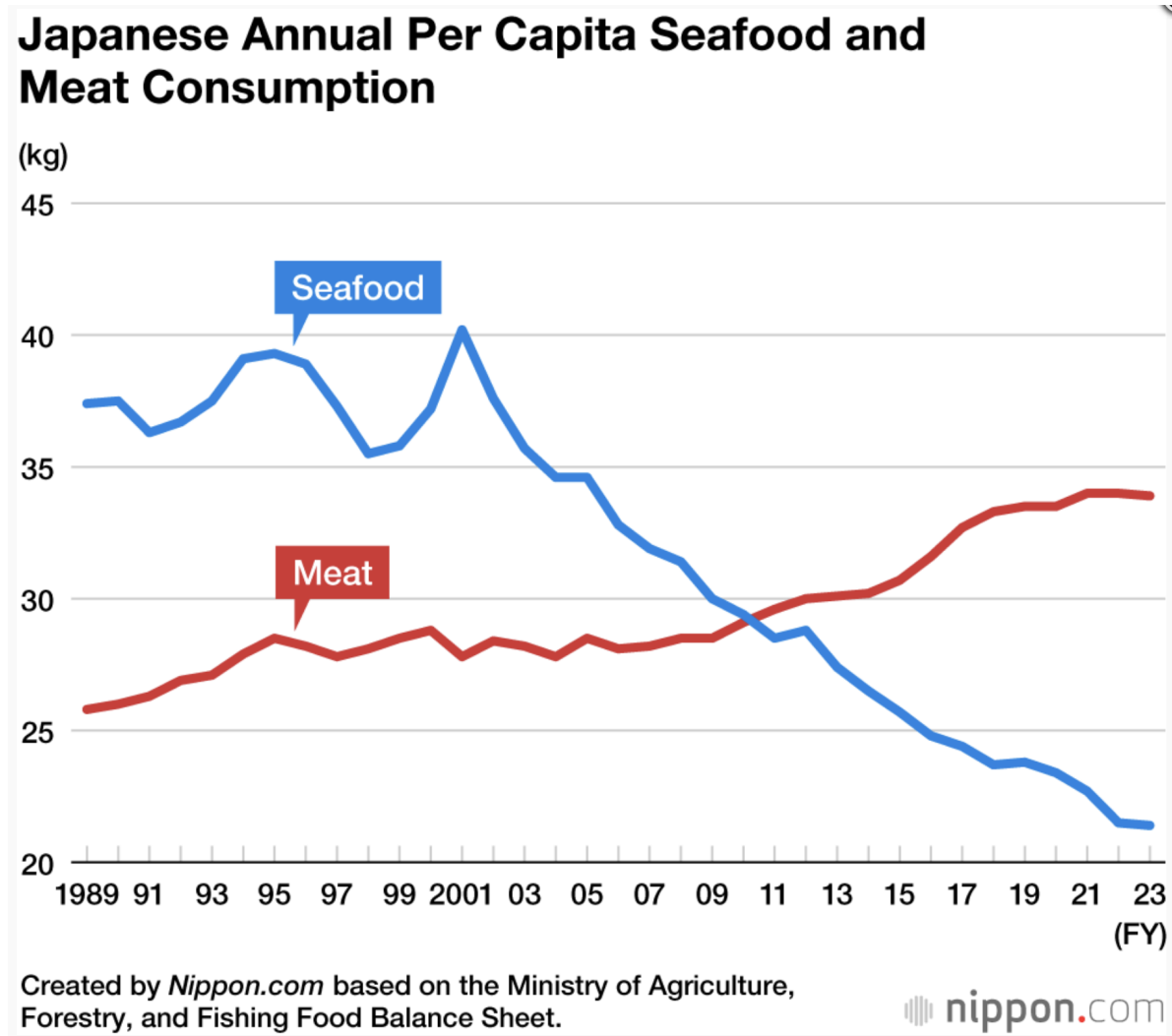


Figure 2: Seafood vs Meat consumption trend in Japan (1989–2023)

Along with a more flexible supply, the shift in consumer demand means that older economic models, which relied on fixed quantities, are no longer accurate. The complex market structure needs empirical verification of previously held assumptions.

3 Data Descriptions

The data is the updated Eales et al. (1997) dataset sourced from the *Family Income and Expenditure Survey* (FIES) published by the Statistics Bureau of Japan which records demand and expenditure for approximately 9,000 households nationwide. The survey reports expenditures and quantities for individual seafood items, which we aggregate into six categories:

1. **HFF:** High-value fresh fish (tuna, sea bream, yellowtail).
2. **MFF:** Medium-value fresh fish (horse mackerel, bonito, flounder, salmon, other raw fish).
3. **LFF:** Low-value fresh fish (sardines, mackerel, saury).
4. **LSC:** Crustaceans (shrimp, lobster, crabs).
5. **CSO:** Cephalopods (cuttlefish, squid, octopus).
6. **SHF:** Shellfish (short neck clams, freshwater clams, oysters, scallops, other shellfish).

Category prices are aggregated using chained Törnqvist indices, normalized to 1.00 in the period of January 1990. Implicit quantities are obtained by dividing each category's monthly expenditure by its price index. The estimation sample runs from January 1985 to November 2025 ($N = 490$).

Table 1 gives the descriptive summary of the dataset. Medium-value fresh fish (MFF) dominates the Japanese seafood basket, with the largest mean budget share at 38.7 percent, followed by high-value fresh fish (HFF) at 23.3 percent. The wide gap between the minimum and maximum budget share in every category reflects the strong seasonality of Japanese seafood consumption as well as market's structural evolution over the decades.

Figures 3 and 4 show the price indices and implicit quantities for the MFF and LSC categories. In each panel the faint line is the raw monthly series and the bold line is its seasonally adjusted,

Table 1: Summary statistics for main variables (1985-2025)

Variable	Mean	Std Dev	Min	Max
Price Index(HFF)	1.051	0.190	0.813	2.028
Price Index(MFF)	0.711	0.191	0.477	1.212
Price Index(LFF)	0.534	0.399	0.113	1.483
Price Index(LSC)	1.186	0.417	0.726	3.078
Price Index(CSO)	0.984	0.225	0.642	1.725
Price Index(SHF)	0.540	0.340	0.114	1.462
Expenditure(HFF)	1,022.831	333.705	599.000	2,601.000
Expenditure(MFF)	1,665.122	427.174	1,091.000	3,046.000
Expenditure(LFF)	271.371	128.426	93.000	749.000
Expenditure(LSC)	596.418	424.420	213.000	2,513.000
Expenditure(CSO)	428.659	173.613	184.000	1,008.000
Expenditure(SHF)	433.098	180.245	131.000	1,090.000
Budget share HFF	0.233	0.020	0.171	0.276
Budget share MFF	0.386	0.049	0.238	0.495
Budget share LFF	0.064	0.030	0.019	0.181
Budget share LSC	0.125	0.047	0.069	0.317
Budget share CSO	0.096	0.018	0.050	0.160
Budget share SHF	0.097	0.020	0.043	0.140

trend-decomposed counterpart. The series are strongly cyclical, and the quantity of both categories trends over the sample; the much higher LSC price index reflects a large increase in crustacean prices relative to 1990. The remaining categories display a mix of positive and negative trends. To accommodate this seasonality, all demand specifications include monthly dummies and are estimated in first differences, as detailed in the following section.

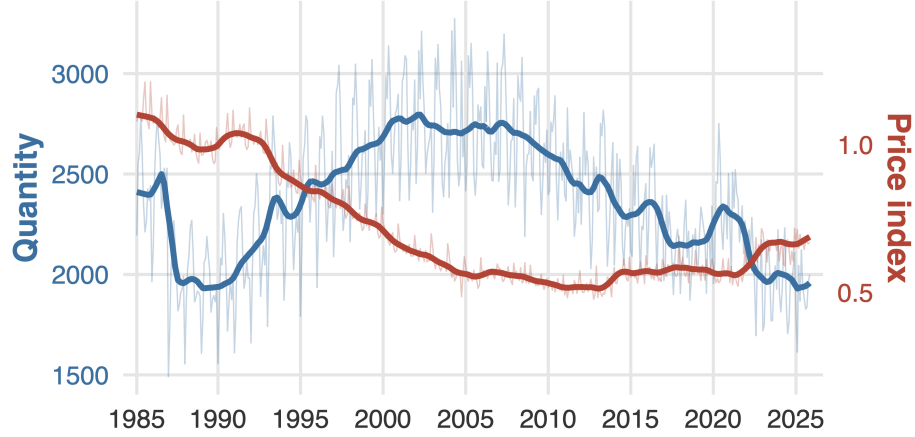


Figure 3: STL trends of Medium-Value Fresh Fish (bold) over observed series (faint)

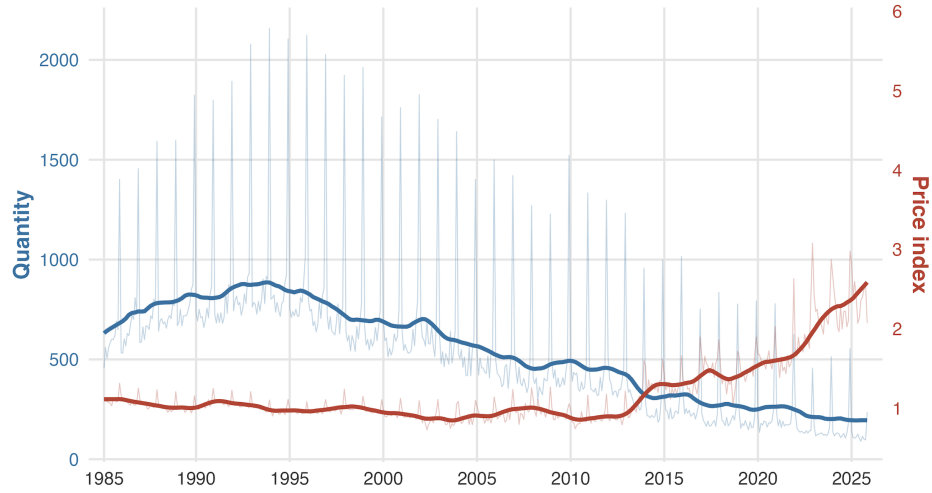


Figure 4: STL trends of Lobster, Shrimp, Cuttelfish (bold) over observed series (faint)

4 Empirical Model

4.1 Generalized Demand Models

Barten (1993) showed the Rotterdam, the differential AIDS, CBS and NBR models can be nested within a generalized ordinary demand systems. Brown et al. (1995) developed a generalized inverse demand system. While these authors parameterized the demand models with Rotterdam dependent variables, for the ease of comparison of the inverse and ordinary demand systems, Eales et al. (1997)

illustrated how both these systems can be reparameterized to have AIDS dependent variables. The generalized inverse demand reparameterized with inverse AIDS dependent variable is:

$$dw_i = \left(\phi_i + \theta_1^I w_i \right) d \ln Q + \sum_{j=1}^N \left[\phi_{ij} + \theta_2^I w_i (\delta_{ij} - w_j) \right] d \ln q_j, \quad (1)$$

where the ϕ 's are coefficients, and θ_1^I and θ_2^I are nesting parameters for the inverse system

Furthermore, the generalized ordinary demand with AIDS dependent variables is reparameterized as:

$$dw_i = \left(\sigma_i + \theta_1^O w_i \right) d \ln Q + \sum_{j=1}^N \left[\sigma_{ij} + \theta_2^O w_i (\delta_{ij} - w_j) \right] d \ln p_j, \quad (2)$$

where the σ 's are coefficients, and θ_1^O and θ_2^O are nesting parameters for the ordinary system.

Restrictions that yield models nested within the generalized demand models are summarized in Table 2. Typical demand restrictions for generalized inverse demand systems include *adding-up*, *homogeneity*, and *symmetry* conditions.

Table 2: Restrictions on the Generalized Models that Yield Alternative Functional Forms

Model	θ_1	θ_2
Inverse AIDS	0	0
Inverse Rotterdam	1	1
Inverse CBS	0	1
Inverse NBR	1	0
AIDS	0	0
Rotterdam	-1	1
CBS	0	1
NBR	-1	0

Note: Notation follows that of equation 1 and 2 in the text.

Adding-up

The **adding-up restriction** ensures that the value of total expenditure is exhausted by the sum of all goods, implying

$$\sum_i \phi_{ij} + \theta_2^I \left(\sum_i w_i \delta_{ij} - w_j \sum_i w_i \right) = 0 \text{ or } \sum_i \phi_{ij} = 0 \quad \text{and}$$

$$\sum_i (\phi_i + \theta_1^I w_i) = 0 \quad \text{or} \quad \sum_i \phi_i = -\theta_1^I$$

Homogeneity

The **homogeneity restriction** arises from the fact that demand functions are homogeneous of degree zero in prices and income:

$$\sum_i \phi_{ij} + \theta_2^I \left(\sum_i w_i \delta_{ij} - w_i \sum_j w_j \right) = 0 \text{ or } \sum_i \phi_{ij} = 0$$

This condition implies that a proportional change in all prices and income leaves demand unchanged.

Symmetry

The **symmetry restriction** results from the symmetry of the substitution matrix:

$$\phi_{ij} = \phi_{ji}$$

Additionally, Antonelli matrix needs to be *negative semidefinite*.

For the **generalized ordinary demand systems** also the restrictions are adding-up, homogeneity and symmetry along with negative semidefiniteness of the Slutsky matrix.

$$\sum_i \sigma_{ij} = 0 \quad \text{and} \quad \sum_i \sigma_i = -\theta_1^O$$

$$\sum_j \sigma_{ij} = 0$$

$$\sigma_{ij} = \sigma_{ji}$$

Elasticities and Flexibilities

Finally, Table 3 provides formulas for the key elasticities and flexibilities associated with the generalized demand systems. These measures are fundamental for interpreting the economic behavior implied by the model.

Table 3: Price and Scale Flexibilities for the Generalized Demand Models

Model	Ordinary	Inverse
Compensated	$e_{ij}^* = \frac{\sigma_{jj}}{w_i} + (\theta_2^O - 1)(\delta_{ij} - w_j)$	$f_{ij}^* = \frac{\phi_{ij}}{w_i} + (\theta_2^I - 1)(\delta_{ij} - w_j)$
Scale flexibility	$e_i = \frac{\sigma_j}{w_i} + \theta_1^O + 1$	$e_i = \frac{\phi_i}{w_i} + \theta_1^I - 1$
Uncompensated	$e_{ij} = \frac{\sigma_{ij} - \sigma_i w_j}{w_i} + (\theta_2^O - 1)(\delta_{ij} - (\theta_1^O + \theta_2^O)w_j)$	$f_{ij} = \frac{\phi_{ij} + \phi_j w_i}{w_i} + (\theta_2^I - 1)(\delta_{ij} + (\theta_1^I - \theta_2^I)w_j)$

4.2 Forecasting Strategy

We use a two-stage procedure: first we generate component forecasts from the individual demand models, then we combine them using a range of weighting schemes.

4.2.1 Component Model Estimation

Following Klaiber and Holt (2010), we first generate component (individual) forecasts from the inverse and ordinary demand systems. Each system is estimated by seemingly unrelated regression (SUR) on a fixed rolling window updating every month and used to generate unconditional one- to four-step-ahead forecasts over 2001-2025; the initial estimation window spans January 1985 to December 2000. To keep the forecasts genuinely unconditional (ex ante), we assume no knowledge

of future explanatory variables: all stochastic right-hand-side variables—total expenditure, and prices for the ordinary models or quantities for the inverse models are forecasted. We implement an AR(p) model with monthly shift dummies, following the Klaiber and Holt (2010) auxiliary-model approach for RHS quantities and prices, adapted to first-differenced series with BIC lag selection. The Divisia volume index is then constructed from the forecast quantities. Repeating this over the rolling window yields a panel of P component forecasts across the T forecast periods.

4.2.2 Forecast Combination Methods

When the sets of forecast contain independent information, the combination can yield improvements (Bates & Granger, 1969). Forecast combination protects against model specification uncertainty when properly applied (Ramsey & Adjemian, 2024). Following the framework of Ramsey and Adjemian (2024) and augmenting their combination methods with NNLS of Weiss et al. (2018), we employ nine combination strategies. Combination weights are estimated on a training set (2001-2017) and evaluated out of sample on a test set (2018-2025) of component forecasts.

Simple Heuristics

1. **Simple Average (SA):** an equal-weighted average of all P component forecasts.
2. **Trimmed Average (TA):** an average that excludes the top and bottom $\lambda\%$ of forecasts to mitigate the impact of outliers.
3. **Inverse Rank (InvR):** weights are assigned proportional to the inverse of the model's rank based on Mean Squared Prediction Error (MSPE) in the training set.

Regression-Based Combinations

4. **Ordinary Least Squares (OLS):** weights are estimated by regressing the realized values y_t on the component forecasts $f_{i,t}$ with an intercept to correct for bias (Aksu & Gunter, 1992).

5. **Non-Negative Least Squares (NNLS):** a constrained version of OLS where weights are restricted to be non-negative ($w_i \geq 0$) for greater stability and interpretability (Aksu & Gunter, 1992).
6. **Complete Subset Regression (CSR):** following Elliott et al. (2013), following Elliott et al. (2013), we estimate every possible linear combination of the component models and average the resulting forecasts, a data-driven form of shrinkage.

Eigenvector Approaches

7. **Eigenvector (EIG):** weights derived from the eigenvectors of the MSPE matrix, exploiting the covariance structure of forecast errors to minimize variance (Hsiao & Wan, 2014).
8. **Bias-Corrected Eigenvector (EIG-BC):** a refinement of EIG that adjusts for systematic bias in the component models.

5 Empirical Results

5.1 Preliminary estimates from generalized demands

Table 4 reports the estimates from the generalized demands calculated at the mean budget shares using the subsample data 2001-2017. 2018-2025 data is held out for the model forecasting performance. Consistent with Eales et al. (1997), we find a dichotomy in the implied market behavior. The ordinary own-price elasticities lie close to -1 for every category, with incomes near $+1$, the pattern of a market in which consumers adjust quantities to hold expenditure roughly fixed. The inverse price flexibilities, by contrast, cluster tightly between -0.33 and -0.81 , well above -1 , implying considerable inelastic demand: when quantities are taken as given, prices must move substantially to clear the market. The divergence between these two sets of estimates confirms that the choice of functional form fundamentally alters the economic interpretation of the market.

Table 4: Generalized Demand Model Elasticity Estimates

Commodity	Flexibility/ Elasticity	Inverse	Ordinary	Mean Share
HFF	Own-price	-0.752 (0.031)	-1.002 (0.049)	0.236
	Scale-income	-0.997 (0.047)	0.980 (0.062)	
MFF	Own-price	-0.818 (0.029)	-0.928 (0.038)	0.382
	Scale-income	-1.063 (0.034)	0.899 (0.042)	
LFF	Own-price	-0.335 (0.034)	-1.255 (0.144)	0.074
	Scale-income	-0.756 (0.099)	1.440 (0.224)	
LSC	Own-price	-0.565 (0.041)	-0.926 (0.074)	0.117
	Scale-income	-0.901 (0.106)	1.279 (0.145)	
CSO	Own-price	-0.655 (0.036)	-1.017 (0.058)	0.091
	Scale-income	-0.988 (0.073)	0.893 (0.093)	
SHF	Own-price	-0.621 (0.038)	-1.034 (0.062)	0.101
	Scale-income	-1.073 (0.090)	0.883 (0.116)	

5.2 Non-Nested Hypothesis Tests

We further conduct Davidson and MacKinnon's p-test to see if a competing model captures important information that the baseline model misses to determine the best fit. We estimate the models using iterated 3SLS with four distinct instrument sets, following Eales et al. (1997). We adhere closely to the instruments proposed by Eales et al. (1997) which include the first and twelfth-order lags of prices, quantities, and expenditures and macro variables. Macrovariables capture the broader structural transition of the Japanese economy: the Yen/USD exchange rate, CPI, cumu-

lated diffusion indexes, the prime interest rate/call rate, government bond yields, household size, disposable income, expenditure and wage rate.

The results in Table 5 differ from Eales et al. (1997), where the inverse model dominated clearly. The inverse rejects the ordinary in all four instrument sets, and the ordinary rejects the inverse in three of four instrument sets. This mutual rejection by both models suggests neither of the model fully captures the market structure. This further suggests that the underlying data-generating process is no longer robustly captured by a single structural form. However this result should be interpreted with caution as the lags and macro instrumental variables used by Eales et al. (1997) have lost their predictive power in the recent times. For every endogenous right-hand-side variable, joint first-stage F-statistics on the combined lag and macro instrument set fall well below the 10 percent maximal-bias critical value of 22.30 with $K = 14$ instruments per equation, with individual F-statistics ranging from 0.11 to 6.06 (Stock & Yogo, 2005). The market has outgrown the old models with the variables that predicted prices in the 1990s unable to explain any longer. This further supports the structural shift hypothesis for the Japanese fish market and warrants our need to reevaluate the models appropriate for the demand estimation but the instruments cannot be used for them being weak.

Table 5: Nonnested Tests of the General Demand Models

Instrument Sets	Null versus Alternative Models	
	Ordinary vs Inverse	Inverse vs Ordinary
First Lags, Twelfth Lags, and Macro Variables	0.268* (0.017)	0.361* (0.092)
First Lags and Twelfth Lags	0.240* (0.016)	0.353* (0.093)
First Lags and Macro Variables	0.233* (0.024)	0.331* (0.126)
Twelfth Lags and Macro Variables	0.254* (0.040)	0.187 (0.147)

Note: Entries in the table are the coefficient estimates, with standard errors in parentheses. An asterisk (*) indicates that the estimate is statistically significant at the 5% level, rejecting the null model.

5.3 Forecast

The identification failure documented in Section 5.2 means the traditional model-selection tools available in the demand-system literature — the non-nested p-test in particular — cannot resolve which specification better describes modern Japanese seafood demand. Rather than continue other structural tests, we turn to out-of-sample forecast performance. If one specification systematically dominates in forecasting terms, that provides an alternative empirical basis for model selection; if no specification dominates, forecast combination offers a way to minimize the model uncertainty rather than committing to any single one.

5.3.1 Static conditional forecasts

We first replicate the static conditional forecasting of Eales et al. (1997) where predictions are generated using parameters estimated from the training set (held fixed) and the real values of the explanatory variables in the test set.

Table 6 reports Theil's U statistics and the ratio of Root Mean Squared Error (RMSE) for these

conditional forecasts. Two key patterns emerge. First, all inverse and ordinary demand models outperform the naive no-change forecast ($U < 1$). Second, consistent with Eales et al. (1997), the Inverse specification outperforms the Ordinary specification in pairwise comparisons. This result, however, is not achievable in real-time forecasting where future quantities and prices are unknown, motivating the unconditional exercise in the next subsection.

Table 6: Comparison of Generalized Models' Out-of-Sample Forecast

Commodity	Theil's U-Statistics		RMSE Ratio (Ordinary/Inverse)
	Inverse	Ordinary	
HFF	0.56	0.80	1.43
MFF	0.20	0.31	1.60
LFF	0.24	0.63	2.63
LSC	0.11	0.18	1.64
CSO	0.34	0.50	1.49
SHF	0.37	0.55	1.51

Note: Using 2018-2025 for out of sample forecasting.

5.3.2 Unconditional Rolling Window Forecast

While conditional forecasts provide a theoretical benchmark, they use the future exogenous variables in the estimation which are unknown in real world. So to simulate the real world scenario, we adopt the unconditional rolling window methodology utilized by Klaiber and Holt (2010). In this method, a moving window of fixed width is firstly selected. Now for each forecast step, the estimation window shifts forward by one month to four months, dropping the oldest observations generating component forecasts. To ensure the forecasts are truly unconditional (ex-ante), we implement an AR(p) model with monthly shift dummies, on auxiliary-model approach for RHS quantities and prices, adapted to first-differenced series with BIC lag selection following the Klaiber and Holt (2010). We then combine these component forecasts using the weights estimated in the training period (2001–2017) and test period (2018-2025) following the framework of Ramsey and Adjemian (2024).

5.4 Forecast Combination

We forecast each of the eight nested demand models—the inverse and ordinary AIDS, CBS, NBR, and Rotterdam specifications—by the unconditional rolling-window procedure explained above, at horizons of one to four months, and combine them using nine methods from Ramsey and Adjemian (2024) (SA, TA, INV, OLS, NNLS, CSR, EIG and EIGBC). Forecasts of the demand systems are evaluated by unweighted system RMSE.

Figure 5 reports the forecast errors of individual models at each horizon and figure 6 reports for combination models. The forecast errors reported are summed across all the six fish categories. Firstly, no single model is best throughout among the individual models: the leading individual model is the inverse AIDS at two and four steps, and the ordinary models at other steps. Upon comparison of all models, combination models have lower forecast errors than best individual models at all the horizon. While simple combination techniques (SA and TA) outperform all other models in the first horizon, this changes as we move on to other horizons. The forecasting performance begins to suffer as we increase the horizon as expected. The result suggests more dramatic forecasting performance of inverse models however the result must be interpreted with caution as the forecasting error on RHS quantities for inverse model is higher compared to that of RHS price of ordinary model. The forecasting error on the auxiliary AR forecasts of quantities exceeds that of prices by roughly a factor of two even after seasonal adjustment, which mechanically favors ordinary specifications under unconditional forecasting. This asymmetry shows that Japan relies heavily on imports to set the domestic prices. While the combination forecasting does improve the forecasting error, the improvement is not very significant most likely because of low pool diversity.

Table 7: Individual Demand forecast accuracy, (2001-2025)

Model	Training RMSE	Test RMSE
Ord_AIDS	0.01743	0.02205
Inv_AIDS	0.01691	0.02206
Ord_CBS	0.01826	0.02269
Ord_Rotterdam	0.02110	0.02539
Ord_NBR	0.02054	0.02597
Inv_NBR	0.02735	0.03208
Inv_CBS	0.03239	0.03458
Inv_Rotterdam	0.04247	0.04811

Table 8: Combined forecast accuracy, (2001-2025)

Model	Training RMSE	Test RMSE
INV	0.01666	0.02131
NNLS	0.01691	0.02169
EIG	0.01717	0.02177
CSR	0.01416	0.02186
OLS	0.01404	0.02223
TA	0.01932	0.02291
SA	0.02011	0.02353
EIGBC	0.02306	0.02642

Now on the composite forecasts, simple and trimmed averages outperform all the models on one-step-ahead forecasts, but this changes as the horizon increases. By four step horizons complex weighted combination models outperform simple averaging.

6 Conclusion

In recent years there has been considerable interest in re-examining whether the household demand specifications continue to describe consumption behavior once the structural conditions under which they were developed have changed. In the case of Japanese seafood, the inverse demand system favored by Eales et al. (1997) was based on the assumption that domestic fish supply was biologically fixed and prices adjusted to clear a market of fresh fish which are predetermined. No prior work was done to examine how the market changes once import rises significantly and seafood consumption falls through the substitution with meat. In this paper we revisit the generalized inverse and ordinary specifications using four decades of monthly Japanese household data spanning a period across which imports rose from roughly fourteen percent of supply to well over a third. We compare the resulting specifications through the non-nested tests of Eales et al. (1997) through unconditional rolling-window forecasting following Klaiber and Holt (2010), and through forecast combination across the pool of nested specifications following Ramsey and Adjemian (2024).

The structural test results present a different picture than previously reported. Where Eales et al. (1997) found the inverse specification clearly preferred by their non-nested p-tests across every instrument set they considered, we find mutual rejection between the ordinary and inverse models in three of four instrument sets. More strikingly, the joint first-stage F-statistics on the lag and macroeconomic instrument set fall below the Stock and Yogo (2005) critical value for weak instruments in every endogenous variable. The instruments that once identified the supply side in Eales et al.'s data have lost most of their explanatory power in ours. This suggests the structural transition since the macroeconomic variables that predicted domestic landings in the 1980s lose

their transmission mechanism once world prices and exchange rates determine the supply available to Japanese households.

Under conditional forecasting the inverse specification retains its accuracy advantage, confirming that Eales et al.'s original conclusion is not overturned when their information assumption is preserved. Under unconditional forecasting no single specification dominates: the inverse AIDS model leads at horizons two and four while ordinary specifications lead at intermediate horizons. For those that prefer a single best model that better describes modern Japanese seafood demand the result shows mixed conclusion.

Combining across the nested models produces forecasts that are better than best individual model. The magnitude of the accuracy gain is modest, because the eight nested specifications produce forecasts that correlate above 0.90 within each family and around 0.88 across the inverse-ordinary divide. This echoes Klaiber and Holt (2010) finding that combining theory-consistent demand systems with one another yields limited gains.

The results reported here are specific to the data and models pool considered. Future research can apply similar procedures in more diverse pool and see if the combination gain increases.

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7 Appendix

Table 9: Individual Demand forecast accuracy in multiple horizons, 2001-2025

Model	Training sample					Test sample				
	$h = 1$	$h = 2$	$h = 3$	$h = 4$	Total	$h = 1$	$h = 2$	$h = 3$	$h = 4$	Total
Ord_AIDS	0.00839	0.00877	0.00888	0.00881	0.01743	0.01030	0.01099	0.01135	0.01142	0.02205
Inv_AIDS	0.00813	0.00852	0.00859	0.00858	0.01691	0.01044	0.01088	0.01140	0.01136	0.02206
Ord_CBS	0.00849	0.00917	0.00952	0.00931	0.01826	0.01021	0.01137	0.01187	0.01185	0.02269
Ord_Rotterdam	0.01073	0.00913	0.01092	0.01129	0.02110	0.01348	0.01139	0.01296	0.01286	0.02539
Ord_NBR	0.01093	0.00895	0.01033	0.01075	0.02054	0.01406	0.01170	0.01315	0.01293	0.02597
Inv_NBR	0.01147	0.01328	0.01463	0.01504	0.02735	0.01306	0.01586	0.01749	0.01736	0.03208
Inv_CBS	0.01193	0.01625	0.01845	0.01738	0.03239	0.01185	0.01791	0.01963	0.01869	0.03458
Inv_Rotterdam	0.01677	0.02094	0.02363	0.02294	0.04247	0.01754	0.02425	0.02710	0.02617	0.04811

Table 10: Combined forecast accuracy in multiple horizons, 2001-2025

Model	Training RMSE					Test RMSE				
	$h = 1$	$h = 2$	$h = 3$	$h = 4$	Total	$h = 1$	$h = 2$	$h = 3$	$h = 4$	Total
INV	0.00825	0.00823	0.00847	0.00837	0.01666	0.01001	0.01055	0.01104	0.01100	0.02131
NNLS	0.00813	0.00824	0.00872	0.00871	0.01691	0.01045	0.01039	0.01118	0.01131	0.02169
EIG	0.00889	0.00816	0.00866	0.00862	0.01717	0.01116	0.01038	0.01100	0.01099	0.02177
CSR	0.00704	0.00709	0.00714	0.00703	0.01416	0.01022	0.01062	0.01153	0.01131	0.02186
OLS	0.00698	0.00705	0.00709	0.00696	0.01404	0.01022	0.01063	0.01193	0.01160	0.02223
TA	0.00848	0.00973	0.01026	0.01007	0.01932	0.00987	0.01157	0.01217	0.01205	0.02291
SA	0.00866	0.01015	0.01077	0.01052	0.02011	0.00988	0.01192	0.01262	0.01244	0.02353
EIGBC	0.00895	0.01167	0.01313	0.01197	0.02306	0.00993	0.01346	0.01497	0.01394	0.02642

Table 11: Individual Demand forecast accuracy in multiple horizons, 2005-2025

Model	Training RMSE					Test RMSE				
	$h = 1$	$h = 2$	$h = 3$	$h = 4$	Total	$h = 1$	$h = 2$	$h = 3$	$h = 4$	Total
Inv_AIDS	0.00920	0.00961	0.00984	0.00993	0.01930	0.01162	0.01207	0.01244	0.01229	0.02422
Ord_AIDS	0.00956	0.00997	0.01014	0.01015	0.01992	0.01131	0.01225	0.01250	0.01245	0.02427
Ord_CBS	0.00963	0.01034	0.01074	0.01067	0.02071	0.01123	0.01285	0.01321	0.01299	0.02519
Ord_Rotterdam	0.01178	0.00971	0.01142	0.01169	0.02236	0.01471	0.01159	0.01286	0.01268	0.02601
Ord_NBR	0.01204	0.00952	0.01085	0.01115	0.02186	0.01528	0.01153	0.01277	0.01253	0.02620
Inv_NBR	0.01252	0.01464	0.01605	0.01662	0.03008	0.01435	0.01773	0.01950	0.01893	0.03548
Inv_CBS	0.01304	0.01761	0.01979	0.01895	0.03508	0.01271	0.01924	0.02084	0.01952	0.03670
Inv_Rotterdam	0.01804	0.02255	0.02531	0.02498	0.04581	0.01866	0.02580	0.02883	0.02717	0.05082

Table 12: Combined forecast accuracy in multiple horizons, 2005-2025

Model	Training RMSE					Test RMSE				
	$h = 1$	$h = 2$	$h = 3$	$h = 4$	Total	$h = 1$	$h = 2$	$h = 3$	$h = 4$	Total
CSR	0.00771	0.00768	0.00791	0.00775	0.01553	0.01055	0.01085	0.01151	0.01133	0.02213
OLS	0.00763	0.00757	0.00782	0.00765	0.01534	0.01038	0.01111	0.01174	0.01153	0.02241
INV	0.00933	0.00926	0.00962	0.00956	0.01888	0.01108	0.01153	0.01188	0.01167	0.02309
EIG	0.01002	0.00908	0.00965	0.00964	0.01921	0.01226	0.01116	0.01161	0.01152	0.02328
NNLS	0.00936	0.00933	0.01034	0.01102	0.02007	0.01168	0.01139	0.01264	0.01328	0.02454
TA	0.00956	0.01091	0.01150	0.01145	0.02176	0.01092	0.01301	0.01351	0.01317	0.02539
SA	0.00973	0.01136	0.01205	0.01194	0.02261	0.01091	0.01341	0.01402	0.01359	0.02608
EIGBC	0.00998	0.01407	0.01457	0.01253	0.02582	0.01098	0.01686	0.01722	0.01427	0.03009

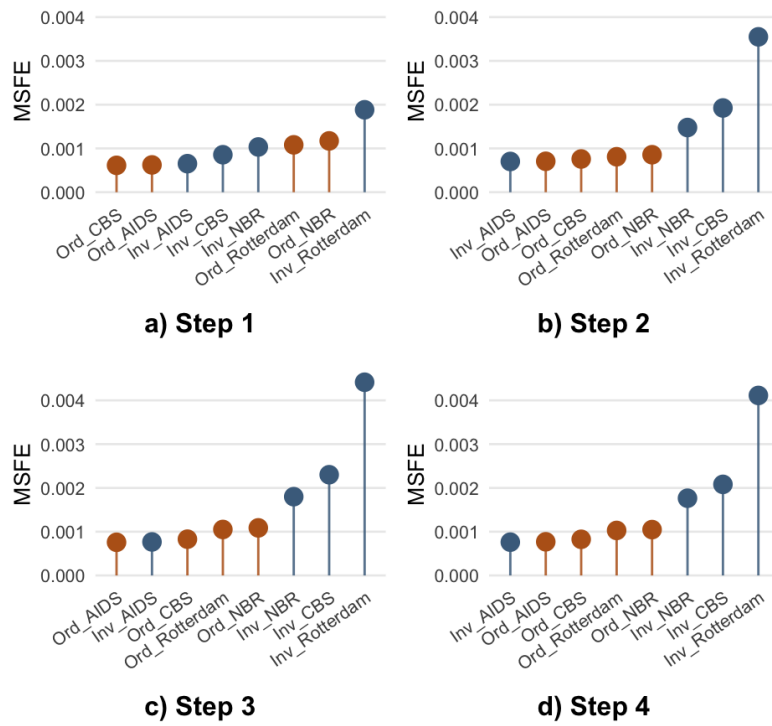


Figure 5: Mean squared forecast errors by step for individual models

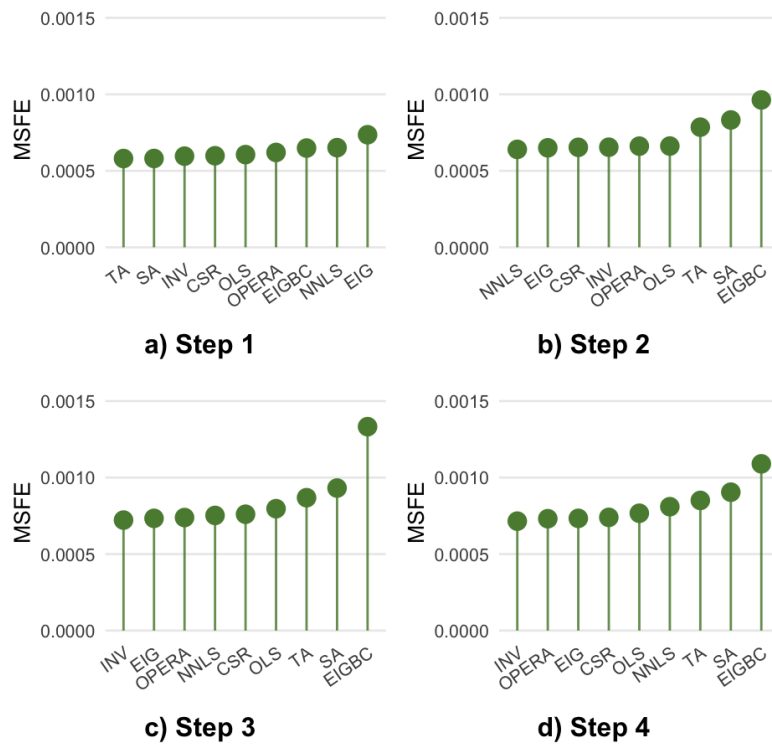


Figure 6: Mean squared forecast errors by step for combination models